



PAPER



CODE

Perceptrons and Localization of Attention's Mean-Field Landscape

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I. Overview

We study the forward pass of a **Transformer** as an interacting particle system on $\mathbb{S}^{d-1}$:
layers $\leftrightarrow$ **time** | **tokens** $\leftrightarrow$ **particles** | **LayerNorm** $\leftrightarrow$ **sphere**.
 For suitable parameters, the mean-field limit is a **Wasserstein gradient flow** on $\mathcal{P}(\mathbb{S}^{d-1})$.

Question. How does adding the *perceptron block* modify the equilibria of *pure attention*?

Answer at a glance. The perceptron *reshapes* the energy landscape and *localizes* the (even diffuse!) equilibria of pure attention into singular, seed-independent distributions.

II. The variational reduction of a Transformer

Roughly, the ℓ -th layer of a post-LN encoder (e.g. BERT) maps the i -th token $x_i^\ell \in \mathbb{R}^d$ as:

DISCRETE DYNAMICAL SYSTEM

$$x_i^\ell \mapsto \text{LN} \left(x_i^\ell + \Delta t V^\ell \sum_{j=1}^n \frac{e^{\langle B^\ell x_i^\ell, x_j^\ell \rangle}}{\sum_k e^{\langle B^\ell x_i^\ell, x_k^\ell \rangle}} x_j^\ell \right) \mapsto \text{LN} (y_i^\ell + \Delta t W^\ell \sigma(A^\ell y_i^\ell + b^\ell))$$

=: y_i^ℓ (ATTENTION) =: $x_i^{\ell+1}$ (PERCEPTRON)

where $\text{LN}(s) := s/\|s\|$, and σ is typically ReLU or GELU.

INTERACTING PARTICLE ODE

$$\dot{x}_i(t) = \mathbf{P}_{x_i(t)}^\perp \left(V(t) \sum_{j=1}^n \frac{e^{\langle B(t)x_i(t), x_j(t) \rangle}}{\sum_k e^{\langle B(t)x_i(t), x_k(t) \rangle}} x_j(t) + W(t) \sigma(A(t)x_i(t) + b(t)) \right)$$

where $\mathbf{P}_x^\perp := I_d - xx^\top$ is the orthogonal projection onto $\mathbb{T}_x \mathbb{S}^{d-1}$.

Deriving the gradient flow

1 Simplifications.

- ▷ **Isotropic, time-independent attention:** $V = \pm I_d$ and $B = \beta I_d$ ($\beta > 0$).
- ▷ **Time-independent perceptron and zero bias:** $(W(t), A(t))$ constant and $b(t) = 0$.

2 Gradient structure.

- ▷ **Unnormalized attention:** Drop the softmax $e^{\langle Bx_i, x_j \rangle} / (\sum_k e^{\langle Bx_i, x_k \rangle}) \mapsto e^{\beta x_i \cdot x_j} / n$.
- ▷ **Conservative perceptron:** $W\sigma(Ax) = \nabla V_\theta(x)/2 \iff W = A^\top \Omega$ (Ω diagonal matrix).
 ◦ Example: $V_\theta(x) = \mathbf{1}^\top \Omega (Ax)_+^2$ for the ReLU perceptron.

3 Mean-field/infinite-context limit.

- ▷ **Token distribution:** $\mu_t^n = \frac{1}{n} \sum_i \delta_{x_i(t)} \xrightarrow{n \rightarrow \infty} \mu_t$ uniformly on $[0, T]$, which solves:

WASSERSTEIN GRADIENT FLOW (ASCENT + / − DESCENT)

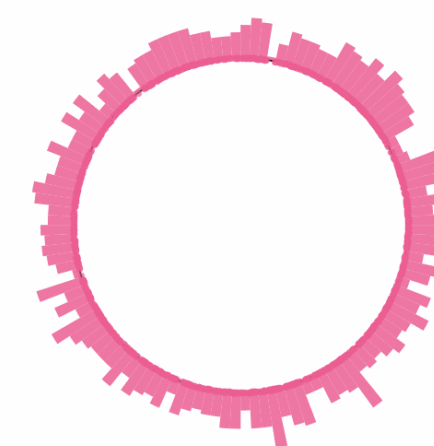
$$\partial_t \mu_t + \text{div} \left(\mu_t \mathbf{P}_x^\perp \left[\pm \int e^{\beta x \cdot y} y d\mu_t(y) + \frac{1}{2} \nabla V_\theta(x) \right] \right) = 0.$$

- **Energy:** $E_{\beta, \theta}[\mu] = \frac{1}{2\beta} \iint e^{\beta x \cdot y} d\mu(x) d\mu(y) \pm \frac{1}{2} \int V_\theta(x) d\mu(x).$
- **Stationary:** $\nabla \frac{\delta E_{\beta, \theta}}{\delta \mu}[\mu] = 0$ on $\text{supp } \mu$ $\left[\frac{\delta E_{\beta, \theta}}{\delta \mu}[\mu](x) = \frac{1}{\beta} \int e^{\beta x \cdot y} d\mu(y) \pm \frac{1}{2} V_\theta(x) \right].$
- **Stable:** $\text{Hess}_\mu E_{\beta, \theta}(\xi, \xi) \geq 0 \quad \forall \xi \in \mathbb{T}_\mu \mathcal{P}(\mathbb{S}^{d-1}).$
- **Strictly stable:** $\text{Hess}_\mu E_{\beta, \theta}(\xi, \xi) \geq \kappa \|\xi\|_{L^2(\mu)}^2$ with $\kappa > 0$ indep. of ξ .

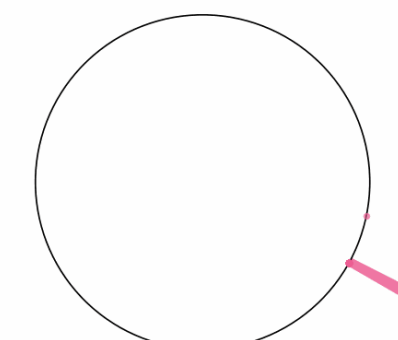
III. Baseline: pure attention

For pure attention, the asymptotics are well understood [2, 1]:

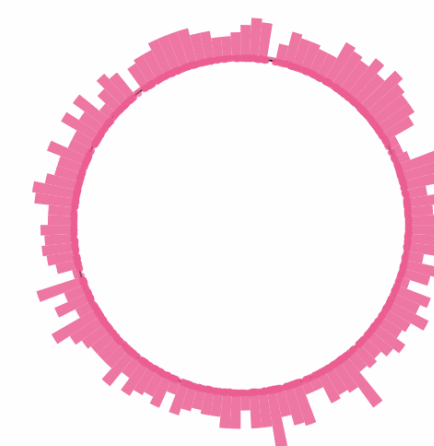
- **Ascent** (attractive): Metastability emerges first, then particles collapse to a single Dirac.
- **Descent** (repulsive): Particles equidistribute (global minimizer is the uniform measure).



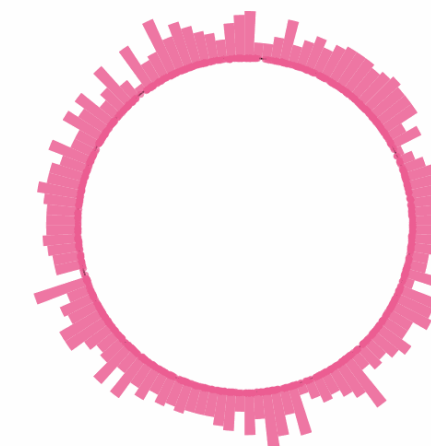
(a) Ascent: initial



(b) Ascent: final



(c) Descent: initial



(d) Descent: final

IV. The perceptron localizes the landscape

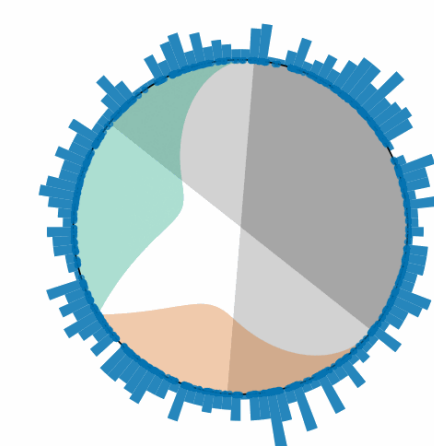
Note: Theorems 1–3 extend to symmetric invertible matrices B , non-zero biases $b \neq 0$ and softmax attention.

Theorem 1 (“Stationarity is rigid enough for the ReLU on the circle”)

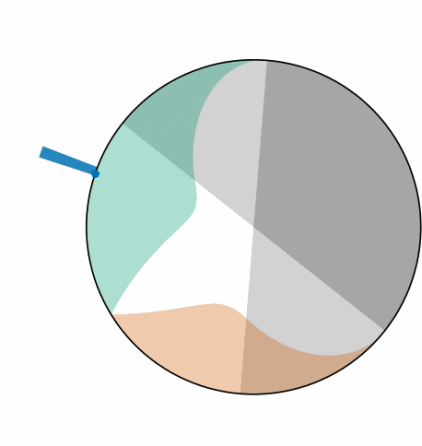
Let $d = 2$, $\beta > 0$, $\sigma(s) = s_+$ and V_θ not real-analytic on $\mathbb{S}^1$. Then every stationary $\mu \in \mathcal{P}(\mathbb{S}^1)$ is purely atomic with finite support.

Proof sketch. 1. On each activation region, stationarity is real-analytic. If $\text{supp}(\mu)$ were infinite, the identity theorem would force $\int e^{\beta x \cdot y} d\mu(y)$ to be a global polynomial.
 2. An invertibility lemma then forces μ to have a polynomial density on $\mathbb{S}^1$.
 3. This propagates real-analyticity globally to V_θ , yielding a contradiction.

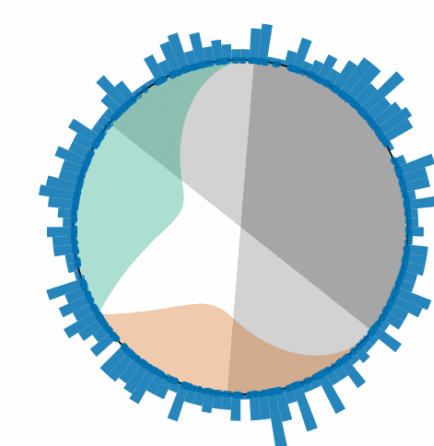
Radial profile: V_θ ■ $V_\theta(x) > 0$ ■ $V_\theta(x) < 0$ ■ $(Ax)_1 \leq 0$



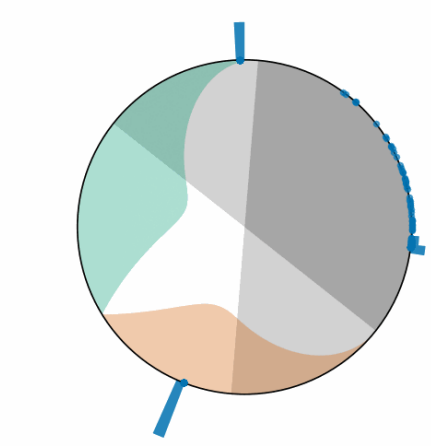
(e) ReLU (Ascent: initial)



(f) ReLU (Ascent: final)



(g) ReLU (Descent: initial)



(h) ReLU (Descent: final)

Theorem 2 (“Under real-analyticity, stationarity is not enough; strict stability collapses the support”)

Let $d = 2$, $\beta > 0$ and V_θ real-analytic (e.g. for σ the GELU). Every strictly stable Wasserstein critical point $\mu \in \mathcal{P}(\mathbb{S}^1)$ is purely atomic with finite support.

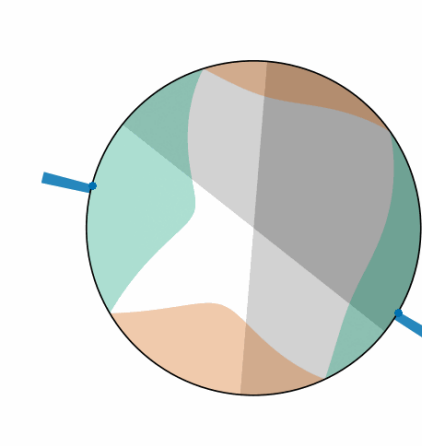
Proof sketch. 1. Infinite support + real-analyticity $\implies$ Wasserstein Hessian at μ becomes:

$$\text{Hess}_\mu E_{\beta, \theta}(\xi, \xi) = -\frac{1}{\beta} \iint K_\beta''(\theta - \phi) \xi(\theta) \xi(\phi) d\mu(\theta) d\mu(\phi) \quad \text{for all } \xi \in \mathbb{T}_\mu \mathcal{P}(\mathbb{S}^1).$$

2. Around any accumulation point, build zero-mean, unit-norm ξ_δ (support size δ) s.t.
 $|\text{Hess}_\mu E_{\beta, \theta}(\xi_\delta, \xi_\delta)| \lesssim \frac{1}{\beta} \sup_{|h| \leq \delta} |K_\beta''(h) - K_\beta''(0)| \xrightarrow{\delta \rightarrow 0} 0$



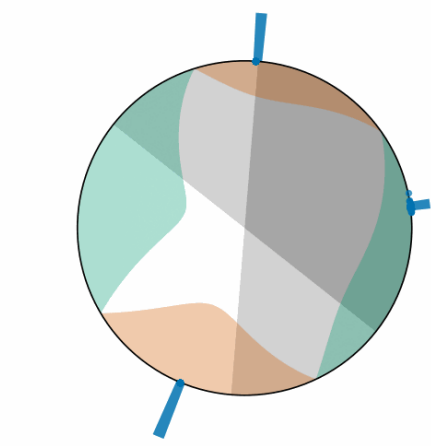
(i) GELU (Ascent: initial)



(j) GELU (Ascent: final)



(k) GELU (Descent: initial)



(l) GELU (Descent: final)

IV. (continued)

Theorem 3 (“Singular support and generic atomicity in every dimension”)

- (a) Let μ satisfy the assumptions of either Theorem 1 or Theorem 2, with $d = 2$ replaced by $d \geq 2$. Then $\text{supp}(\mu)$ has zero surface measure.
 (b) Let $d \geq 2$ and fix μ . For a dense set of (β, θ) , if μ is stationary for (β, θ) then it is purely atomic with finite support for GELU, and purely atomic on $\{x : (Ax)_+ \neq 0\}$ for ReLU.

Proof sketch. (a): Analogous to $d = 2$ (identity theorem for multivariate functions).

(b): For GELU and an open dense set of (β, θ) , the stationarity equation has full-rank differential. By parametric transversality, all zeros of the field are isolated. By compactness, they are finite. For ReLU, this argument applies on every active stratum defined by $(Ax)_+$.

V. Anti-concentration (repulsive regime)

Theorem 4 (“Local concavity prevents a trivial collapse”)

Let $d = 2$, $\text{Lip}(\sigma) = 1$ with $\sigma(0) = 0$. Let $\mu = \sum_{i=1}^N m_i \delta_{\varphi_i}$ be a stable critical point with $N \geq 2$. If a sub-cluster $\mathcal{S} \subseteq \{1, \dots, N\}$ with $|\mathcal{S}| \geq 2$ satisfies

$$\max_{i, j \in \mathcal{S}} \text{dist}_{\mathbb{S}^1}(\varphi_i, \varphi_j) \leq \frac{1}{2\sqrt{\beta}},$$

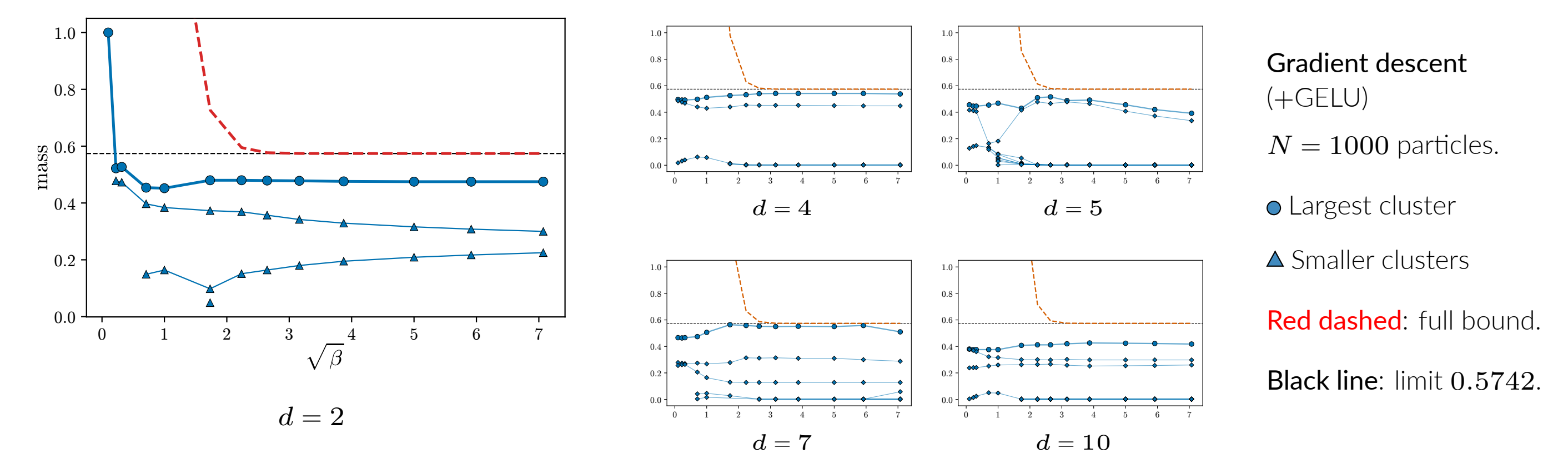
then, for $\beta > 0$ large enough,

$$\sum_{i \in \mathcal{S}} m_i \leq 0.5742 + O(e^{-\beta})$$

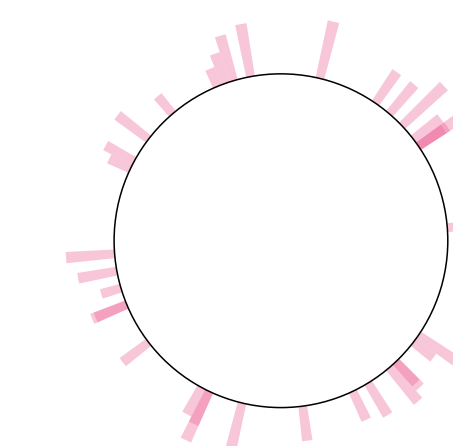
Proof sketch. Test against ξ supported on $\mathcal{S}$ with zero mean, and split the Hessian into:

$$\text{Hess}_\mu E_{\beta, \theta}(\xi, \xi) = \underbrace{\mathcal{T}_{\text{in}}}_{\text{intra-cluster}} + \underbrace{\mathcal{T}_{\text{out}}}_{\text{inter-cluster}} + \underbrace{\mathcal{T}_\theta}_{\text{perceptron}} \geq 0.$$

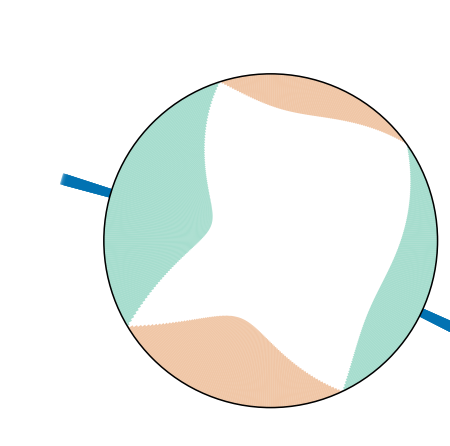
The kernel $e^{\beta \cos \varphi}$ is concave for $|\varphi| \leq (2\sqrt{\beta})^{-1}$, thus $\mathcal{T}_{\text{in}} < 0$, while $\mathcal{T}_{\text{out}}, \mathcal{T}_\theta$ are bounded.



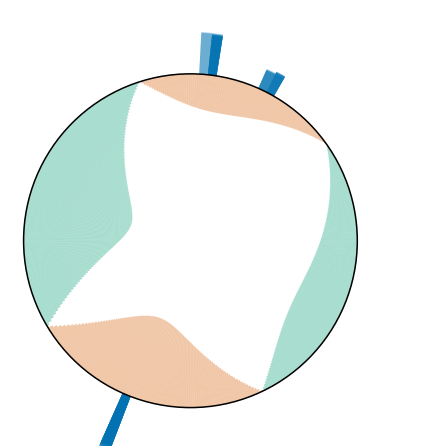
VI. Localization is robust across initializations



(m) Pure att. (ascent) · 10 seeds



(n) + GELU perceptron (ascent)



(o) + GELU perceptron (descent)

References

- [1] B. Geshkovski, C. Letrouit, Y. Polyanskiy, P. Rigollet. *The emergence of clusters in self-attention dynamics*. NeurIPS, 2023.
 [2] B. Geshkovski, C. Letrouit, Y. Polyanskiy, P. Rigollet. *A mathematical perspective on Transformers*. Bull. Amer. Math. Soc. **62**(3):427–479, 2025.